

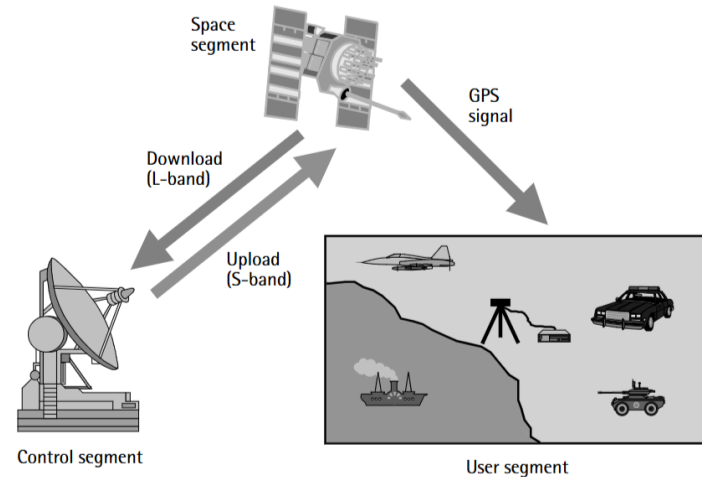
GPS MEO System



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What is GPS?

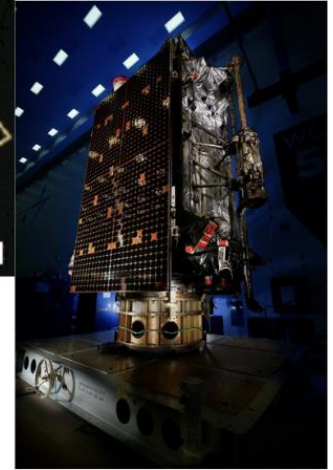
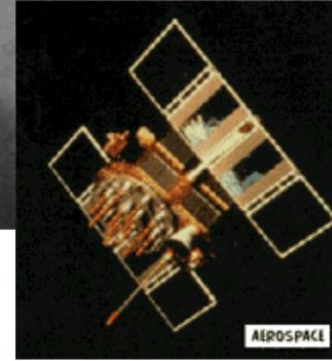
- Medium Earth Orbit (MEO) Global Positioning System
- Uses include navigation, agriculture, surveying and military
- Constellation:
 - At least 24 active satellites
 - Up to 32 satellites
 - Currently 31 active satellites
- Positioning, Navigation, and Timing (PNT)
 - Requires ≥ 4 satellites for accurate positioning (trilateration + clock correction)



GPS Overview [1]

History of GPS

- Block I (1978 - 1985)
 - Prototype system, unencrypted
 - 10 Retired, 1 Failed Launch
- Block II (1989 - 2016)
 - First operational satellites
 - Improved lifespan & added encryption
 - 22 Operational, 7 in Testing/Reserve, 31 Retired, 1 Failed Launch,
- Block III (2018 - 2026)
 - L1C signal, 15-year lifespan
 - 8× improved anti-jamming
 - 9 Operational, 1 in Testing/Reserve

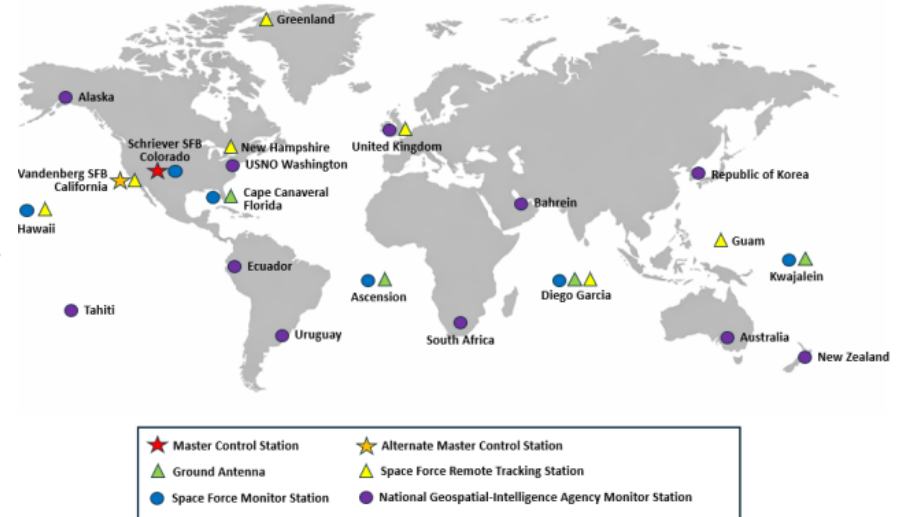


Photos of Block I, Block II, & Block III Satellites [2],[3]

Subsystems

- AOCS
 - Employ three axis stabilization using magnetic torquers [9].
 - Use thrusters for station keeping.
- TTC&M
 - Satellites tracked by worldwide network of ground stations
 - Uploads orbit, clock, atmospheric, and other data through S-band link
- Power
 - 4480 W of power at the end of 15 year design life
- Communication
 - Downlink L-band between 1.18 and 1.61 GHz to civilian (L1, L2, L5, L1C) and military channels

GPS Control Segment



Link Budget - Devices



AN/TRC-202 GPS Transportable Ground Antenna[4]

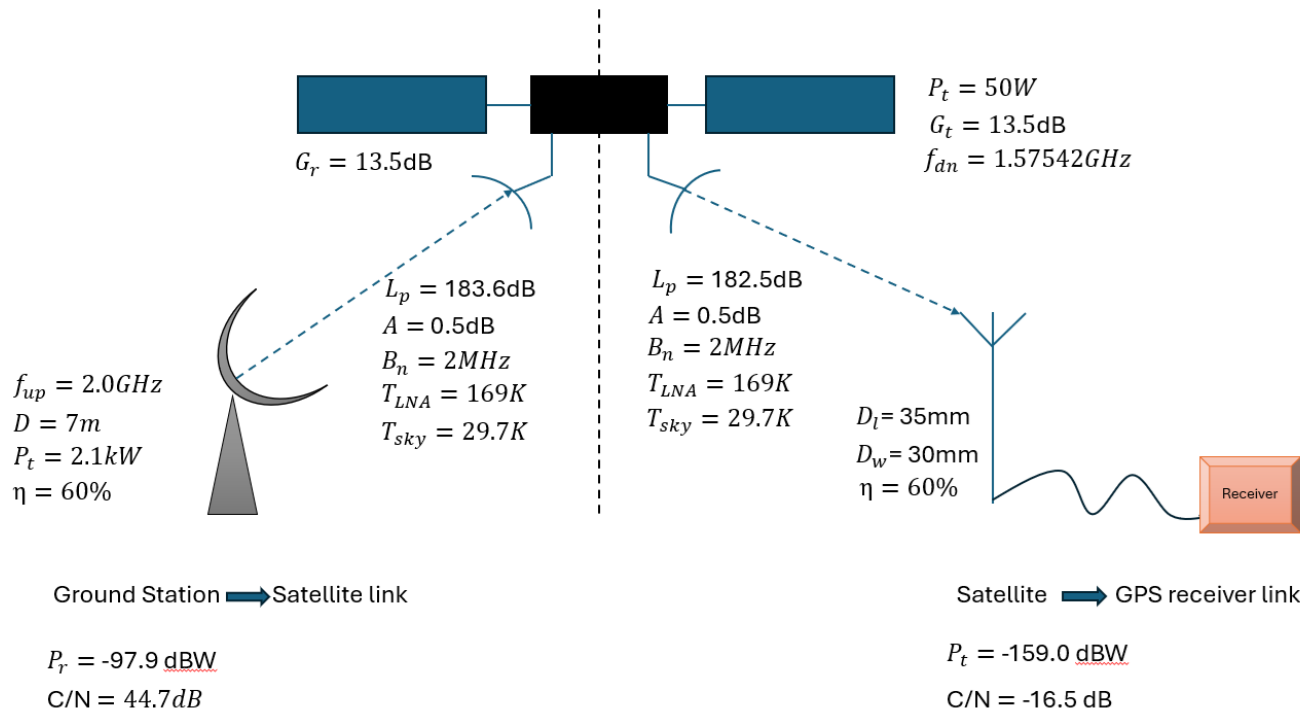
- Created in 1990 as a backup if any of the 4 worldwide GPS Antennas experienced an outage.
- Antenna: 7m in diameter
- Transmit Power: 2.1kW
- Uplink frequency: 2.0GHz



Bingfu Car GPS Antenna[5]

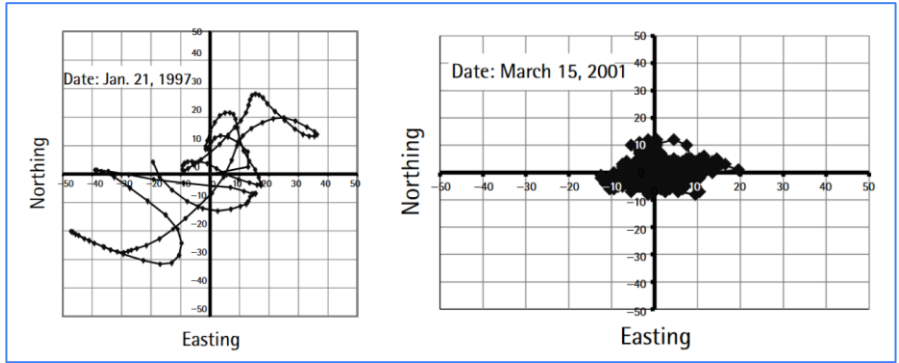
- Useful for built-in vehicle GPS systems and other GPS receivers.
- Antenna: 35mm X 30mm
- NF: 2.0 dB

Link Budget



Errors and Accuracy

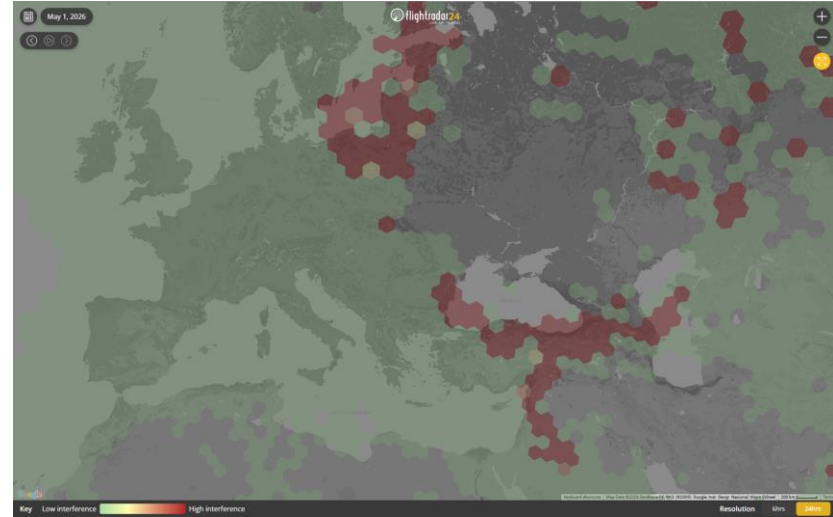
- Selective availability [1]
 - Intentional degrading of civilian GPS service by the U.S. government
 - 1990 - 2000
 - Delta errors - dithered satellite clock
 - Epsilon errors - incorrect orbital data
- Other sources of error [6]:
 - Ionospheric delay: 1.3 m
 - Satellite clock errors: 1.43 m
 - Ephemeris errors: 0.57 m
 - Multipath: 1.0 m
 - Receiver noise: 1.4 m
 - **2DRMS Error (95%): 2-3 m**



Position variation of GPS receiver with (left) and without (right) selective availability [1]

GPS Jamming

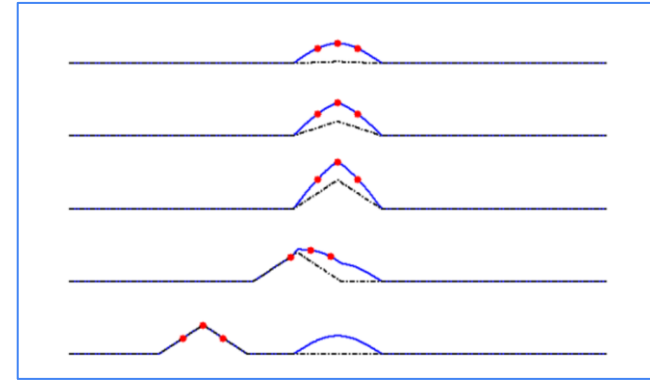
- Radio interference that introduces excess noise into receiver and prevents accurate demodulation of GPS code signal [6]
- Unintentional
 - Inadvertent emissions
 - Solution through regulation and enforcement
- Intentional
 - Has played a major role in military conflict zones (Ukraine, Middle East)
 - Takes advantage of weak GPS signal at receiver
 - Impacts to commercial flight navigation
- Antenna nulling can be used as defense but often difficult to mitigate



GPS interference reported by aircraft in a 24 hour period [7]

GPS Spoofing [8]

- Deceive GPS receiver into asserting false position
- Example attack:
 - 1. Transmit false code at lower power than real code
 - 2. Raise power of false code to exceed real code
 - 3. Vary position/timing information in false code to deceive receiver
- Defenses use encryption (M-code) or signal processing techniques
- Advanced attacks (meaconing, nulling) can counter defenses



Blue = Real GPS Signal
Black = Spoofing GPS Signal
Red = Receiver Track

Future of GPS

- GPS IIIF satellites
 - 22 satellites planned for 2027–2037
 - 60× anti-jamming capability
 - Expected full Block III constellation by 2037
- Continued market growth for consumer receivers



Photo of GPS Block III in development [2]

Questions?

References

- [1] A. El-Rabbany, *Introduction to GPS: The Global Positioning System.*, 2nd Ed., Artech House, 2002.[Accessed: April 6, 2026]
- [2] Lockheed Martin, “From a Rocket to Your Pocket: Behind the GPS III Evolution,” 2026.
- [3] NASA Jet Propulsion Laboratory, “GPS Quick Look,” Jet Propulsion Laboratory (JPL), California Institute of Technology. Accessed: May 3, 2026. [Online]. Available: <https://space.jpl.nasa.gov/msl/QuickLooks/gps1QL.html>
- [4] “GPS TRANSPORTABLE GROUND ANTENNA EXHIBIT,” ccspacemuseum.org, 2025. <https://ccspacemuseum.org/artifacts/gps-transportable-ground-antenna/>.
- [5] “Bingfu Car GPS Antenna SMA Male Waterproof Active GPS Navigation Antenna for Car Stereo Radio Head Unit GPS Navigation System Modem Vehicle Tracker Real Time Monitor Security Camera IoT Router Gateway,” Bingfushop, 2026. <https://bingfushop.com/products/bingfu-car-gps-antenna-sma-male-waterproof-active-gps-navigation-antenna-for-car-stereo-radio-head-unit-gps-navigation-system-modem-vehicle-tracker-real-time-monitor-security-camera-iot-router-gateway?srltid=AfmBOor-wOXgBWqT2eEUuRWEiSVFNiHUB1eUFgVcamEYkVCN6r0SuQJD>.
- [6] T. Pratt and J. Allnutt, “Satellite and the Global Positioning System,” in *Satellite Communications*, 3rd ed. Hoboken, NJ, USA: John Wiley & Sons Ltd, 2020, ch. 12.
- [7] “GPS jamming map.” flightradar24.com. Accessed: Apr. 30, 2026. [Online]. Available: <https://www.flightradar24.com/data/gps-jamming>
- [8] M. L. Psiaki and T. E. Humphreys, "GNSS Spoofing and Detection," in Proceedings of the IEEE, vol. 104, no. 6, pp. 1258-1270, June 2016, doi: 10.1109/JPROC.2016.2526658.
- [9] O. Samir, “Attitude Determination and Control of GPS Satellites: Stabilization, Orbital Insertion, and Operational Control Mechanisms,” The University of Texas at Arlington, Aug. 3, 2025, doi: 10.48550/arXiv.2508.01660. Available: [2508.01660] Attitude Determination and Control of GPS Satellites: Stabilization, Orbital Insertion, and Operational Control Mechanisms. [Accessed: April 6, 2026]